

MS Algebra Recitation 6

Neil Makur

1 Recall

Recall the following:

Proposition (Frattini Argument). *If G is a finite group, $N \trianglelefteq G$, and $P \in \text{Syl}_p(N)$, then $G = N_G(P)N$.*

As an example of how to use this result, we have the following:

Proposition. *Let G be a finite group where all maximal (proper) subgroups are normal. Then every Sylow subgroup of G is normal. In particular, G is isomorphic to the direct product of its Sylow subgroups.*

Proof. We only need to show every Sylow subgroup of G is normal, since we know the second implication from last recitation. Let $P \in \text{Syl}_p(G)$, and suppose that $P \not\trianglelefteq G$, so that $N_G(P) < G$. Let M be a maximal subgroup of G containing $N_G(P)$. Then M is normal in G . Since $|M|$ divides $|G|$, P is also a Sylow p -subgroup of M , and so the Frattini argument gives $G = N_G(P)M$. But the latter is contained in M , a contradiction. ■

As another follow up from last time, recall that every subgroup of an abelian group is normal. In particular, the Sylow subgroups are normal, and so we see that any abelian group is isomorphic to the direct product of its Sylow subgroups. We actually have the following theorem, though we won't prove it now.

Proposition. *Any finite abelian p -group is isomorphic to a direct product of cyclic p -groups.*

This proposition then tells us that any abelian group is isomorphic to a direct product of cyclic groups. This is known as the *Fundamental Theorem of Finite Abelian Groups*.

2 Chief Series

Recall that a series of subgroups

$$G = G_n \geq G_{n-1} \geq \cdots \geq G_1 \geq G_0 = 1$$

is said to be *subnormal* if $G_i \trianglelefteq G_{i+1}$ for all i , and *normal* if $G_i \trianglelefteq G$ for all i . We know that a maximal subnormal series (i.e. one that we cannot insert more groups into) is a composition series, and that every quotient must be simple. We'll investigate what a maximal normal series looks like.

Definition. Let G be a group. A maximal normal series of G is called a *chief series*.

What do the quotients of a chief series look like? For

$$G = G_n \geq G_{n-1} \geq \cdots \geq G_1 \geq G_0 = 1$$

to be a chief series, we need that we can never find some $G_{i+1} > G' > G_i$ with $G' \trianglelefteq G$. Quotienting out by G_i , this means we can never find $G_{i+1}/G_i > G'/G_i > 1$ with $G'/G_i \trianglelefteq G/G_i$. That is, the quotient G_{i+1}/G_i must be minimal among normal subgroups of G/G_i . So, we've reduced our question to asking what minimal normal subgroups look like.

Proposition. *Let G be a group, and $N \trianglelefteq G$ be a minimal normal subgroup. Suppose further that N is finite. Then there is some simple group S and some $n \in \mathbb{N}$ such that $N \cong S^n$.*

Proof. Let $S \trianglelefteq N$ be minimal (note that we don't need $S = N$, since S need not be normal in G !). Since N is finite, we can actually find such an S .

We'll consider the smallest normal subgroup of G containing S . This is just given by $\langle gSg^{-1} : g \in G \rangle$. Since this is normal in G and contained in N , it must be equal to N by the minimality of N . Since N is finite, we actually only need finitely many of these. So take $g_1, \dots, g_n$ so that $N = \langle g_i S g_i^{-1} : 1 \leq i \leq n \rangle$. Further, suppose that this is a minimal generating set (analogous to a basis). Write $S_i = g_i S g_i^{-1}$.

Since $S \trianglelefteq N$, we also have that $S_i \trianglelefteq N$ for each i . In particular, by homework, $S_i S_j = S_j S_i$ for each i, j . That is, we can effectively "commute" elements of S_i and S_j . Since $N = \langle S_i : 1 \leq i \leq n \rangle$, repeatedly applying this then gives us that $N = S_1 \cdots S_n$.

We'll now show that $S_i \cap S_1 \cdots S_{i-1} S_{i+1} \cdots S_n = 1$. Without loss of generality, take $i = 1$. Then $S_1 \cap S_2 \cdots S_n$ is normal in N , as it is the intersection of two normal subgroups of N . The minimality of S_1 then tells us that either $S_1 \cap S_2 \cdots S_n = 1$ or $S_1 \cap S_2 \cdots S_n = S_1$. In the former case, we are done, and in the latter, we see that $N = \langle S_i : 2 \leq i \leq n \rangle$, contradicting our definition of the g_i . Thus, $S_1 \cap S_2 \cdots S_n = 1$.

So, we see that $N \cong S_1 \times \cdots \times S_n \cong S^n$, where we use that all S_i are conjugate and hence isomorphic. Finally, we need to show that S (or equivalently S_1) is simple. Let $M \trianglelefteq S_1$. Then

$$M \times 1 \times \cdots \times 1 \trianglelefteq S_1 \times S_2 \times \cdots \times S_n$$

so the minimality of S_1 tells us that $M = 1$ or $M = S_1$, as desired. ■

We end with some useful existence theorems.

Proposition. *Let G be a finite group. Then G has a chief series.*

Proof. Exercise. Similar to the proof that any finite group has a composition series. ■

Proposition. *Let G be a finite group and $N \trianglelefteq G$. Then there exists a chief series of G where one of the terms is N .*

Proof. Let $G = G_n/N \geq G_{n-1}/N \geq \cdots \geq G_1/N \geq G_0/N = N/N$ be a chief series of G/N and let $N = N_m \geq N_{m-1} \geq \cdots \geq N_1 \geq N_0 = 1$ be a chief series of N . Then

$$G = G_n \geq G_{n-1} \geq \cdots \geq G_1 \geq G_0 = N = N_m \geq N_{m-1} \geq \cdots \geq N_1 \geq N_0 = 1$$

is a chief series for G . ■

3 Complementing Groups

Recall that, by homework, whenever $G = HN$ with $N \trianglelefteq G$ and $H \cap N = 1$, we can figure out the structure of G . Let's look into a specific case where this scenario can arise.

Proposition. *Let G be a finite group, $H, K \leq G$ with $|K| = [G : H]$, and suppose that $|H|$ and $|K|$ are relatively prime. Then $G = HK$ and $H \cap K = 1$.*

Proof. We have that $|H \cap K|$ divides both $|H|$ and $|K|$, and so divides $\gcd(|H|, |K|) = 1$. Thus, $H \cap K = 1$. Additionally,

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = |H| \cdot |K| = |H| \cdot [G : H] = |G|.$$

So, given $N \trianglelefteq G$ with $\gcd(|N|, [G : N]) = 1$, if we can find some $H \leq G$ of size $|H| = [G : N]$ then we can use our developed theory to classify the structure of G . We call such a group H a *complement* of N . So we now turn to finding complements.

Theorem (Schur-Zassenhaus). *Let G be a finite solvable group and let $N \trianglelefteq G$ with $\gcd(|N|, [G : N]) = 1$. Then there is some $H \leq G$ complementing N .*

Proof. We'll induct on $|G|$, with $|G| = 1$ being trivial.

Take a chief series

$$G = G_n \geq G_{n-1} \geq \cdots \geq M \geq N \geq \cdots \geq 1$$

of G . Then $M/N \cong S^n$ for some simple group S . So, S appears as a quotient in a composition series of G . Since G is solvable, the Jordan-Hölder theorem tells us that $S \cong \mathbb{Z}/p\mathbb{Z}$ for some prime p (recall by homework that the only abelian simple groups are cyclic of prime order). Let P be a Sylow p -subgroup of M .

First, note that $p \mid [M : N] \mid [G : N]$. Since $|N|$ is relatively prime to $[G : N]$, we see that $p \nmid |N|$. In particular, $P \cap N = 1$. Thus,

$$|PN| = \frac{|P| \cdot |N|}{|P \cap N|} = |P| \cdot |N| = [M : N] \cdot |N| = |M|,$$

and so $M = PN$. Set $X = N_G(P)$.

Suppose first that $X < G$. Since $M \trianglelefteq G$ is a term in a chief series, the Frattini argument tells us that $G = XM$. Thus,

$$G = XM = X(PN) = (XP)N = XN.$$

Now, taking orders and rearranging gives

$$[G : N] = [X : X \cap N].$$

Suppose otherwise that $X = G$, so that $P \trianglelefteq G$. Consider NP/P , the image of N under quotienting by P . We have that $NP/P \trianglelefteq G/P$ (since $N, P \trianglelefteq G$ so that $NP \trianglelefteq G$). So, by induction, we can find $H/P \leq G/P$ with $|H/P| = [G/P : NP/P] = [G : NP]$. Then,

$$|H| = |P| \cdot [G : NP] = \frac{|P| \cdot |N \cap P|}{|N| \cdot |P|} \cdot |G| = \frac{|G|}{|N|} = [G : N],$$

and so we are done. ■

Note that complements are by no means unique, since the only thing we know about them is their order. For instance, if H complements N , then so does H^g for all $g \in G$. Remarkably, it turns out that *all* complements are of this form. That is, for N as in the theorem, all complements of N are conjugate to each other. This theorem also holds in all groups, though the proof is significantly more complicated.